

DAY 13

Question 2

The Guarded Pivot

(Medium, Arrays)

You are given an array `arr` of `N` integers (can include negative numbers and duplicates).

Find the smallest index `i` such that:

1. `arr[i]` is strictly greater than every other element in the array (i.e., it is the unique maximum), AND
2. The sum of all elements to its left equals the sum of all elements to its right: `sum(arr[0..i-1]) == sum(arr[i+1..N-1])`

If no such index exists, return -1.

Constraints:

$1 \leq N \leq 10^5$

$-10^4 \leq arr[i] \leq 10^4$

Expected time complexity: $O(N)$

Expected auxiliary space: $O(1)$ (excluding input array)

Examples:

Input: `arr = [5, 3, 9, 3, 5]`

Output: 2

Explanation: `arr[2] = 9` is the unique max.

Left sum = $5+3 = 8$, Right sum = $3+5 = 8$. Equal $\rightarrow$ index 2 is the "guarded pivot".

Input: `arr = [1, 2, 3]`

Output: -1

Explanation: `arr[2] = 3` is the unique max, but left sum = 3, right sum = 0. Not equal.

Input: `arr = [7, 7, 2, 1]`

Output: -1

Explanation: No unique maximum exists (7 appears twice), so no valid pivot is possible.